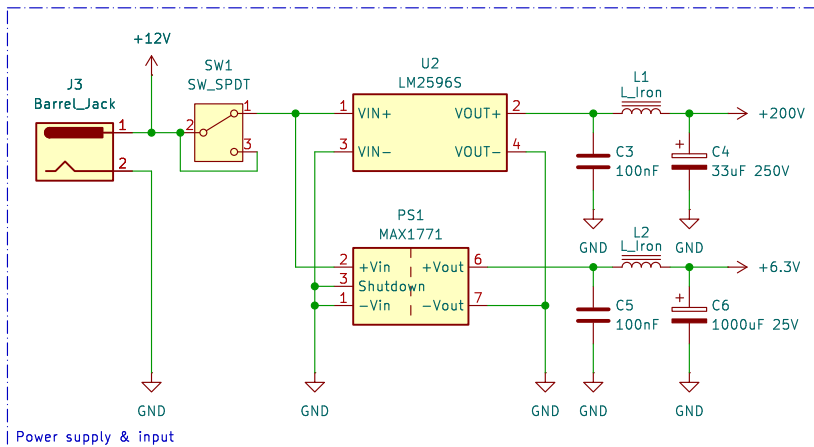
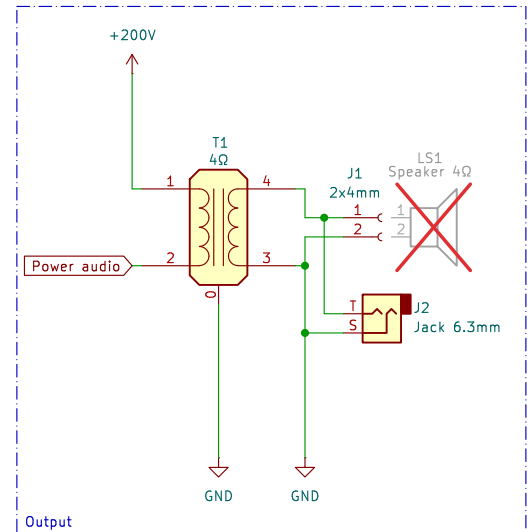
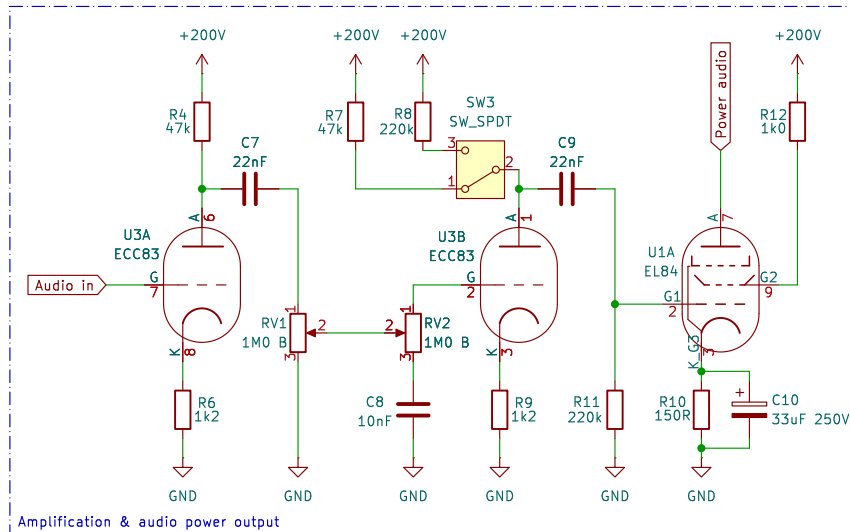
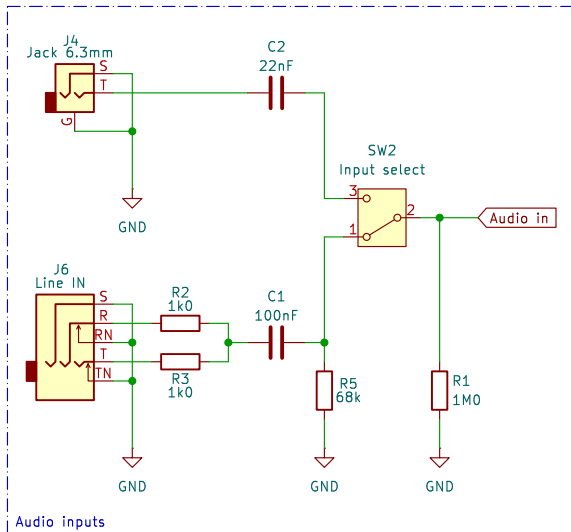




Output transformer
 - Primary $Z = 5.8k\Omega$ $R = 172.3\Omega$
 - Secondary $Z = 4\Omega$ $R = 0.7\Omega$
 Note: Z is the actual impedance of the transformer winding, while R is the DC resistance

EL84 Calculations
 Desired current: 40mA
 - $R_c = 6V / 0.04A = 150\Omega$
 - $U_a = 200V - (172.3\Omega * 0.04A) = 193.1V$
 - $R_{g2} = \text{at least } 1k\Omega$
 Datasheet: <https://www.tube-data.com/sheets/183/e/EL84.pdf>
 ECC83 (1st. stage)
 Desired current: 1.2mA
 - $R_a = (200V - 150V) / 0.0012A = 41.7k\Omega$ - we will use 47kΩ
 - $R_c = 1.2k\Omega$
 - $U_a = 150V$

